

InCCA Software Automatic Bidirectional LIS Connectivity

Authorization

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Table of Changes

Revision	Date	Description
01	20 Apr 2010	Baseline
02	10 Feb 2011	Modified to match Software version 2.06.01 (final release)

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1. INTRODUCTION

This document describes how to interface the InCCA software with a LIS. It applies to Software versions 2.6.1 or higher.

Computer running LIS and computer running InCCA software must be connected to the same LAN.

LIS sends test requests to InCCA software by writing CSV formatted files in a specified folder. Software automatically loads them and keeps test requests in the Entered Samples frame in the UI. Detailed description for input process is given in Section 4.

Once samples are shown in the Entered Samples frame, User runs tests either selecting “Operations → Samples to Tray” or “Operations → Samples to Tray (Barcode)” if available.

System automatically writes results to CSV formatted output files. LIS must read results from these files. Detailed description for output process is given in Section 5.

Windows Registry keys and values are defined in Section 3.

2. DEFINITIONS, ACRONYMS AND ABBREVIATIONS

The list of terms, acronyms and abbreviations with their associated meanings shown below are used in this document.

- LIS - Laboratory Information System
- PC - Personal Computer
- LAN - Local Area Network
- CSV - Comma Separated Values file format
- UI - User Interface

3. WINDOWS REGISTRY KEYS AND VALUES

Software sets the following Registry key during installation:

HKEY_LOCAL_MACHINE\SOFTWARE\DICONEX\InCCa

A sub-key (HOSTQ) must be added to this key by using “regedit.exe”:

HKEY_LOCAL_MACHINE\SOFTWARE\DICONEX\InCCa\HOSTQ

A set of 12 string values must be written inside this key (recommended values are shown between parenthesis):

DONTDELETEINPUT (0)

INCCALIBRATOR (1)
INCCONTROL (1)
INCPATIENT (1)
INCTECHNICAL (1)
INPUTFILTER (*.csv)
INPUTPATH (input)
OUTPUTINTERVAL ()
OUTPUTPATH (output)
OUTPUTPREFIX (hostq_)
ENABLEINPUT (0)
ENABLEOUTPUT (0)

An additional string value is written the first time the software inputs a file:

INPUTDATE

Specific explanations about how to program this Registry values are shown in Section 4 and 5.

4. INPUT PROCESS

4.1. ENABLING AUTOMATIC INPUT

To enable automatic file input reading, the Registry value ENABLEINPUT must be set to 1 before initiating the InCCA Software session.

4.2. INPUT FILES FORMAT AND STRUCTURE

CSV Input files structure is compatible with previous InCCA Software versions. The following list shows all fields to be written per sample according to the sample type.

for Patients:

PAC, ID, Patient name, Patient surname, Age, SEX*, Doctor, Bed, STAT*,
PID, comment, Method1, Method2, ..., MethodN

for Controls:

CON, Control name, STAT*, Method1, Method2, ..., MethodN

for Calibrators:

CAL, Calibrator Name, STAT*, Method1, Method2, ..., MethodN

*SEX: one character (F or M)

*STAT: 0=Normal 1=STAT

Two “\n” characters must be added at the end of the file as End Of File indicator.

4.3. INPUT FILES FOLDER

The file folder on which the InCCA Software will read input files is defined in the registry value INPUTPATH. If the default value isn't modified, default folder will be used:

C:\Documents and Settings\[User]\input

4.4. INPUT FILES MANAGEMENT

InCCA Software checks input folder every minute. If the InCCA Software fails to find the input folder, automatic loading of tests will be disabled for this session. InCCA Software won't create the input folder; therefore this task must be done as a system setup task during bidirectional interface setup.

All valid files in the input folder will be read at once by the InCCA Software. Valid files must be created according to Section 4.2 but only newer files than the last file imported by the Software will be imported.

Once file/s is/are read, InCCA Software writes value INPUTDATE in the Registry the date and time of the last created/modified file. Files created or modified before this date and time won't be read again. The first time the Software imports a file, the value INPUTDATE is created and written.

If the Registry value DONTDELETEINPUT is set to 0, imported files will be deleted from the input folder. If the Registry value DONTDELETEINPUT is set to 1, imported files will remain in the input folder.

To use a file extension different than default, the Registry value INPUTFILTER can be modified to the desired file extension.

5. OUTPUT PROCESS

5.1. ENABLING AUTOMATIC OUTPUT

To enable automatic output files generation, the Registry value ENABLEOUTPUT must be set to 1 before initiating the InCCA Software session.

5.2. OUTPUT FILES FORMAT AND STRUCTURE

Output files maintain the same structure as in older InCCA Software versions for the main group of variables according to the sample type. Each sample type can be included or not in the exported file by setting its related Registry value to 1 (enabled) or 0 (disabled)

for Patients (Registry value INCPATIENT)

Process number, Order number, method's name, PID, Patient's name, Patient's surname, Age, Sex, ID, Doctor, Bed, Report, Blank (1 blank, 0 determination), Number of re-dilutions

for Controls (Registry value INCCONTROL)

Process number, Order number, Method's name, Control name, "CON", "", "", "", "", "", Report, Blank (1 blank, 0 determination), Number of re-dilutions

for Calibrators (Registry value INCCALIBRATOR)

Process Number, Order number, Method's name, Calibrator Name, "CAL", "", "", "", "", "", Report, Blank (1 blank, 0 determination), Number of re-dilutions

Technical information exportation has been changed related to previous versions. If the Registry value INCTECHNICAL is set to 0, no technical information will be exported; if it's set to 1, technical information will be appended to the sample line in the exported file. The following fields will be added:

Concentration, Maximum Value, Minimum Value, Unit, Bichromatic (1 bichromatic, 0 monochromatic)

5.3. OUTPUT FILES FOLDER

The file folder on which the InCCA Software will write output files is defined in the registry value OUTPUTPATH. If the default value isn't modified, default folder will be used:

C:\Documents and Settings\[User]\output

The folder will be created the first time the InCCA Software writes a file.

5.4. OUTPUT FILES MANAGEMENT

The Registry value OUTPUTINTERVAL specifies the frequency in minutes of the output files to be written. Output files will be created when a test is finished; it will remain open until the time specified in OUTPUTINTERVAL elapses, moment in which the file will be closed and a new file will be created if necessary.

Output files name will begin with the string set in the Registry value OUTPUTPREFIX and it will continue with the date and time of the first test written to the file. An example of valid output file name is

hostq_2011-02-10T10_57_07.csv